XAMS optical simulation: model, validation and first results

Geant4 optical simulation of the XAMS TPC with the meshes as built — geometry v3 from the drawings · 5 October 2026 · optical_sim/ (local git), user guide: optical_sim/docs (MkDocs)

Summary.

• Question: is the low S2 light on the top PMT a geometric effect? **Partly.** Geometry and optics give an S2 top fraction of 0.53–0.58 over all plausible optical parameters and liquid levels. The data without a source come close (0.50–0.68 depending on S2 size, 0.55 at 10–20 kPE); the Cs-137 and Na-22 values (0.27–0.40) are far below anything geometric — they are the load effect of the top PMT, not geometry.

• **The geometry is right:** the top-sector light pattern versus position (asymmetries u, v) lies inside the narrow simulation band over ±24 mm, identically for every source.

• **Fit:** the shapes of S2 versus r, S1 versus depth and the sector pattern are reproduced with two efficiency factors on the top/bottom ratio, ε_{S2} = 1.23 and ε_{S1} = 0.79 (single scatters; QE ratio and top-PMT small-signal loss). Low steel reflectivity is preferred; the other optical parameters are not constrained.

• The S2 area fraction top in the data dips at the centre (r < 5 mm); the model does not, and an insensitive cross between the top-PMT anodes cannot produce it (tested up to 6 mm).

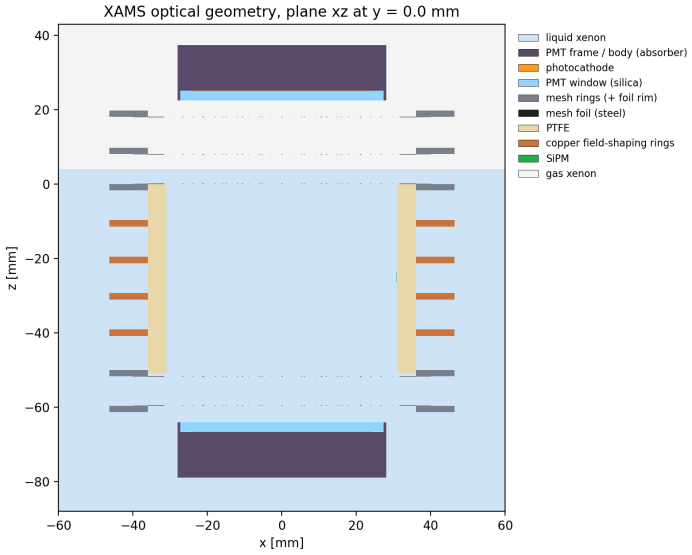
• Only ~23 % of the S2 photons reach a photocathode (QE 0.33 → 7.6 % detected); the mesh foils take ~35 %, ~32 % go sideways into the rings or out of the model. The liquid level (2 → 4 mm) hardly matters.

• Design study: PTFE cylinders also in the gate–anode and anode–top-screen gaps raise the S2 light collection by +37 %, with the cathode–bottom-screen gap +43 % (S1 +28 %) — a factor ~1.4, not more.

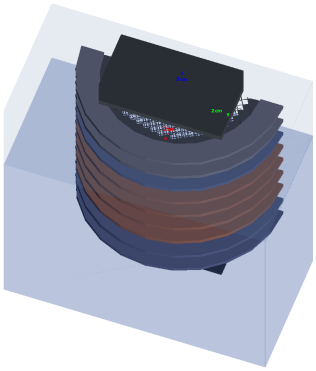
1. Model

One event = one 178 nm photon, started uniformly in the source volume with a random direction and polarisation and followed until it is detected or lost (G4OpticalPhysics: absorption, Rayleigh scattering, boundaries; no scintillation). S2: liquid surface to just below the anode foil, r < 30 mm; S1: cathode to gate, r < 30 mm. Every photon is recorded (origin, end point, how it ended, sensor, reflections); maps are made by binning the origins.

part	model (geometry v2, from the drawings)
meshes (5)	solid 0.15 mm steel foils (Ø80) with 456 square holes 2.45 mm (pitch 2.6 mm) etched in a staircase-shaped area (gate drawing; outer hole corners at r = 32.8 mm), the rest solid; all at 29.4° to the anodes. z: top screen +18, anode +8, gate 0, cathode −51.8, bottom screen −59.6 mm
mesh rings	steel, Ø72–92.6 mm, 1.8 mm assembly; the foil (Ø80) on one face, the ring body on the other side: below the gate and bottom screen, above the cathode, anode and top screen. The gate–anode gap is therefore open sideways between r = 36 and 46 mm
PTFE cylinder	Ø62/72 mm, 50.85 mm, in the bores of the gate and cathode rings (0.8 mm gap to the cathode foil); R 0.90–0.99, specular fraction 0–0.2, rest Lambertian
copper rings	four field-shaping rings Ø72–92.6 × 1.8 mm around the PTFE (8 mm gaps), R 0.2–0.4
xenon	liquid n = 1.69 up to the level (2 / 3 / 4 mm above the gate), Rayleigh 30–60 cm, absorption 1–10 m; gas n = 1.00 above
PMTs	two R12699: silica window 54.6 mm square, 2.5 mm, n = 1.60; photocathode 48.5 mm square with QE 0.33 (top: 4 sectors = ch1–ch4); frame and body absorbing
SiPMs	two, on the PTFE inner wall at z = −25 mm, azimuth 0° / 90°
outside	everything beyond |x|, |y| = 60 mm, below −90 or above +45 mm absorbs (to be made realistic)



Cross section at y = 0 from Geant4's own navigator (exactly what the simulation uses).



3-D view, cut open: PTFE (cream), mesh rings (grey), copper rings, PMTs.

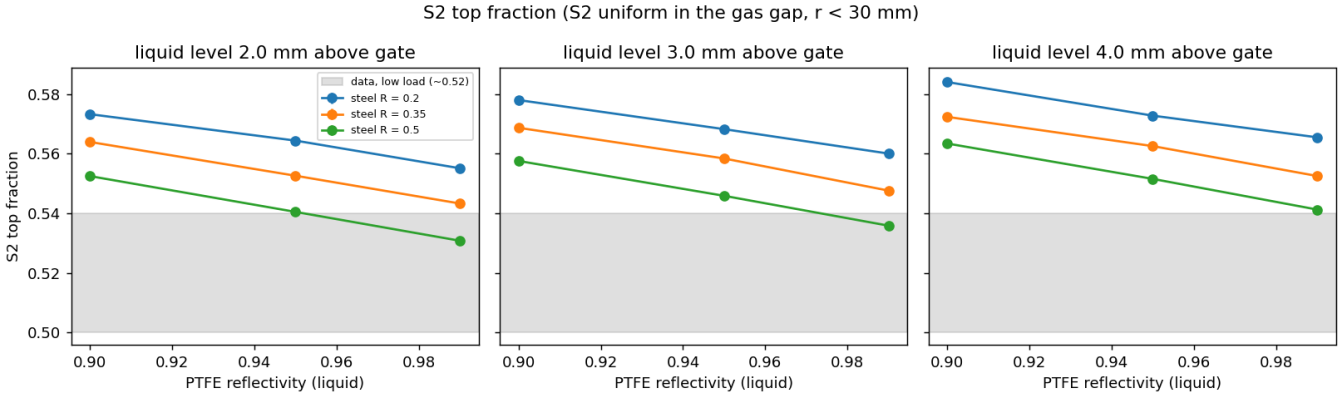
2. Validation

Each implementation step ended with a check against an analytic result:

check	result
solid angle (point source, 48.5 mm cathode at 10/30/60 mm)	detected fraction = solid angle, pulls ≤ 1.1
Fresnel + total internal reflection at the liquid surface, 0–80°	matches (Rs+Rp)/2; TIR from the critical angle 36.3°
mesh transparency, θ 0–75°, φ 0/29.4/45°	Geant4 = analytic ray trace photon by photon ($3 \cdot 10^6$ photons); 0.888 at normal incidence
optical surfaces (PTFE, steel)	reflected fraction = R, specular share = f, diffuse part Lambertian; R = 1 closes
sources and bookkeeping	z ranges follow the liquid level; every photon counted once
seed / statistics	independent halves of every scan point agree (worst pull 2.3)

3. Results

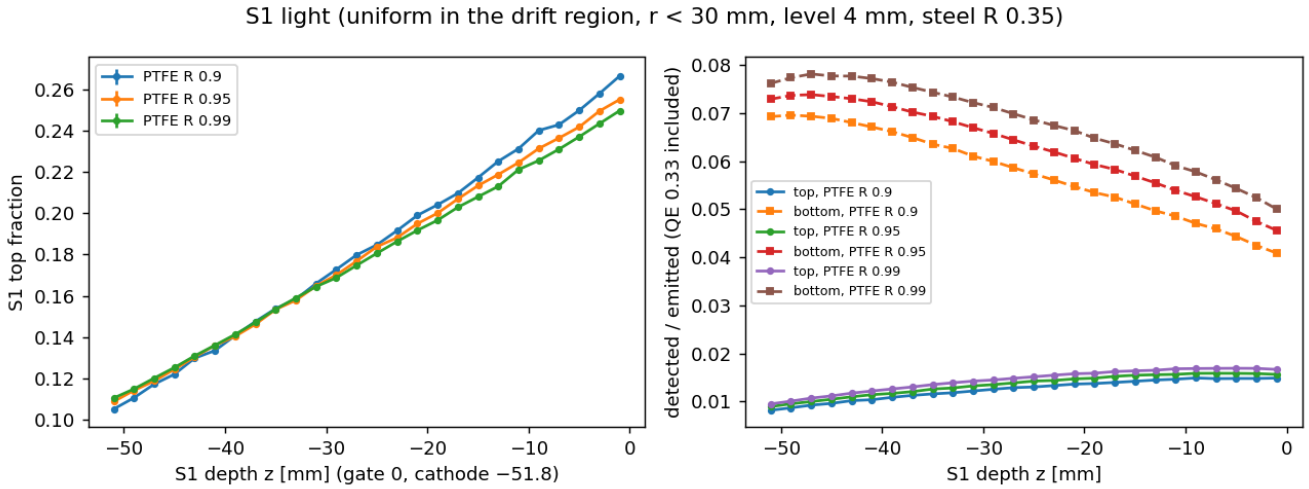
S2 top fraction. Over the full ranges of the free parameters and liquid levels: 0.531–0.584. Strongest knobs: PTFE specular fraction (−0.034 from 0 to 0.2), steel and PTFE reflectivity (≈ −0.02 each over their ranges), Rayleigh length (−0.014); liquid level 2 → 4 mm only +0.010. Versus radius 0.62 at the centre → 0.52 at the edge.



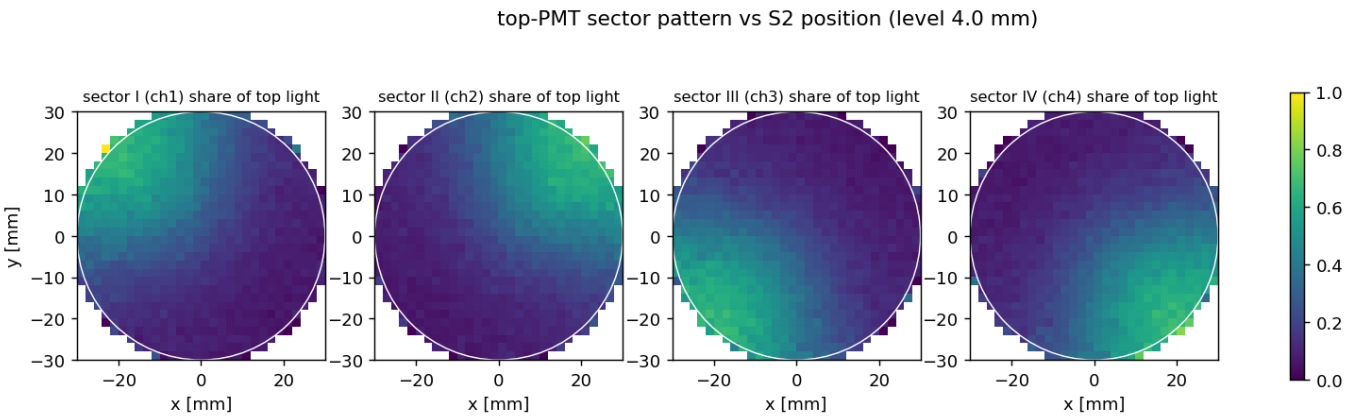
S2 top fraction vs PTFE reflectivity for three steel reflectivities and three liquid levels (10^7 photons per point; statistical errors ±0.0006).

Where the light ends. Reference point: level 4 mm, PTFE R 0.95, steel R 0.35, copper R 0.3.

	S2	S1
detected top / bottom PMT (QE incl.)	4.3 % / 3.3 %	1.3 % / 6.3 %
photocathode reached, not converted (1 − QE)	15.4 %	15.5 %
leaves the TPC region sideways	21.9 %	11.3 %
mesh rings + foil rims	10.3 %	8.9 %
mesh foils: anode / gate / top screen / cathode / bottom screen	12.3 % / 9.7 % / 6.0 % / 3.7 % / 3.5 %	2.6 % / 15.7 % / 1.6 % / 10.5 % / 7.8 %
PTFE	2.6 %	9.0 %
PMT frame / body	6.1 %	6.8 %
copper field-shaping rings	< 0.1 %	< 0.1 %
absorbed in the liquid	0.8 %	2.2 %



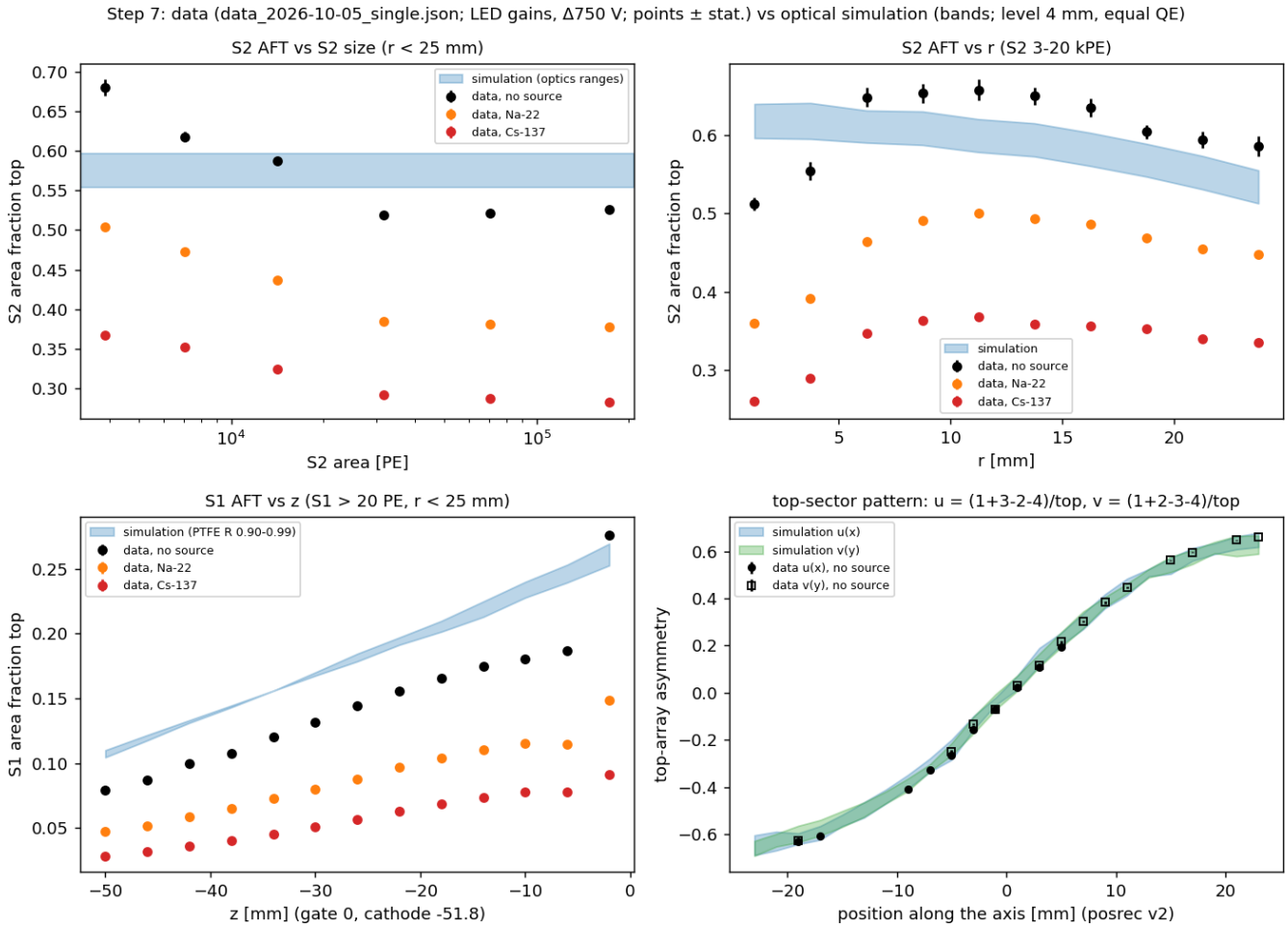
S1: top fraction (0.11 at the cathode → 0.25–0.27 at the gate) and light collection vs depth.



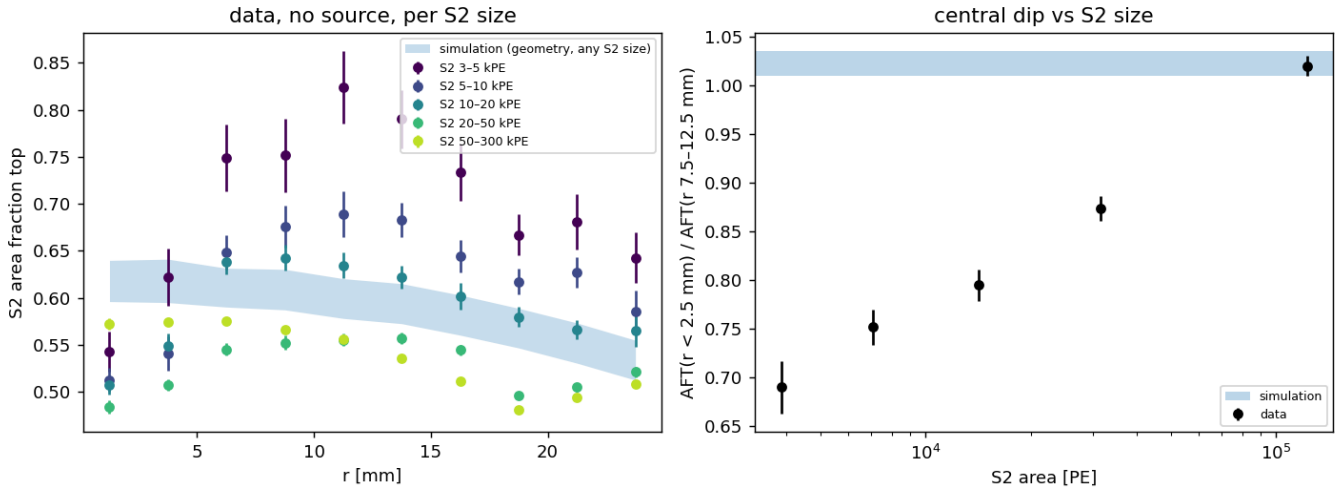
Share of each top-PMT sector in the top light versus S2 position (simulation frame: sector I at −x+y, II +x+y, III −x−y, IV +x−y).

4. Comparison with the data

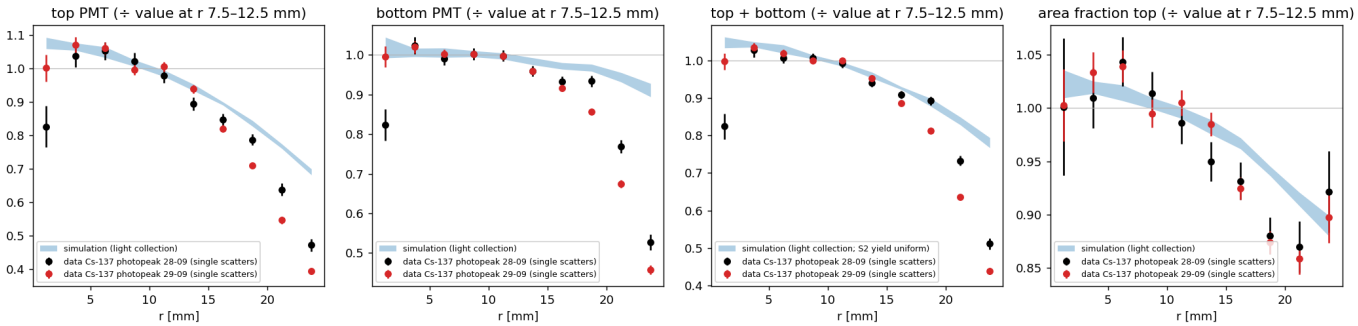
Data: Δ750 V runs of 28/29-09 (central processing v4), no source / Na-22 / Cs-137, **single scatters** (no alternative S2; 20–25 % of the events — a loose cut, alternative S2 < 5 % of the main one, gives the same), r < 25 mm, area fractions with the LED gains only (the v2 load correction f_top undone). Simulation band: all optics points at the 4 mm level, equal QE top and bottom.



Data (points, statistical errors) vs simulation (bands: all optics settings): S2 area fraction top vs S2 size and vs r, S1 vs z, and the top-sector asymmetries u(x), v(y).



Central dip of the S2 area fraction top versus S2 size (data, no source): per size class versus r (left), and centre / ring ratio (right).

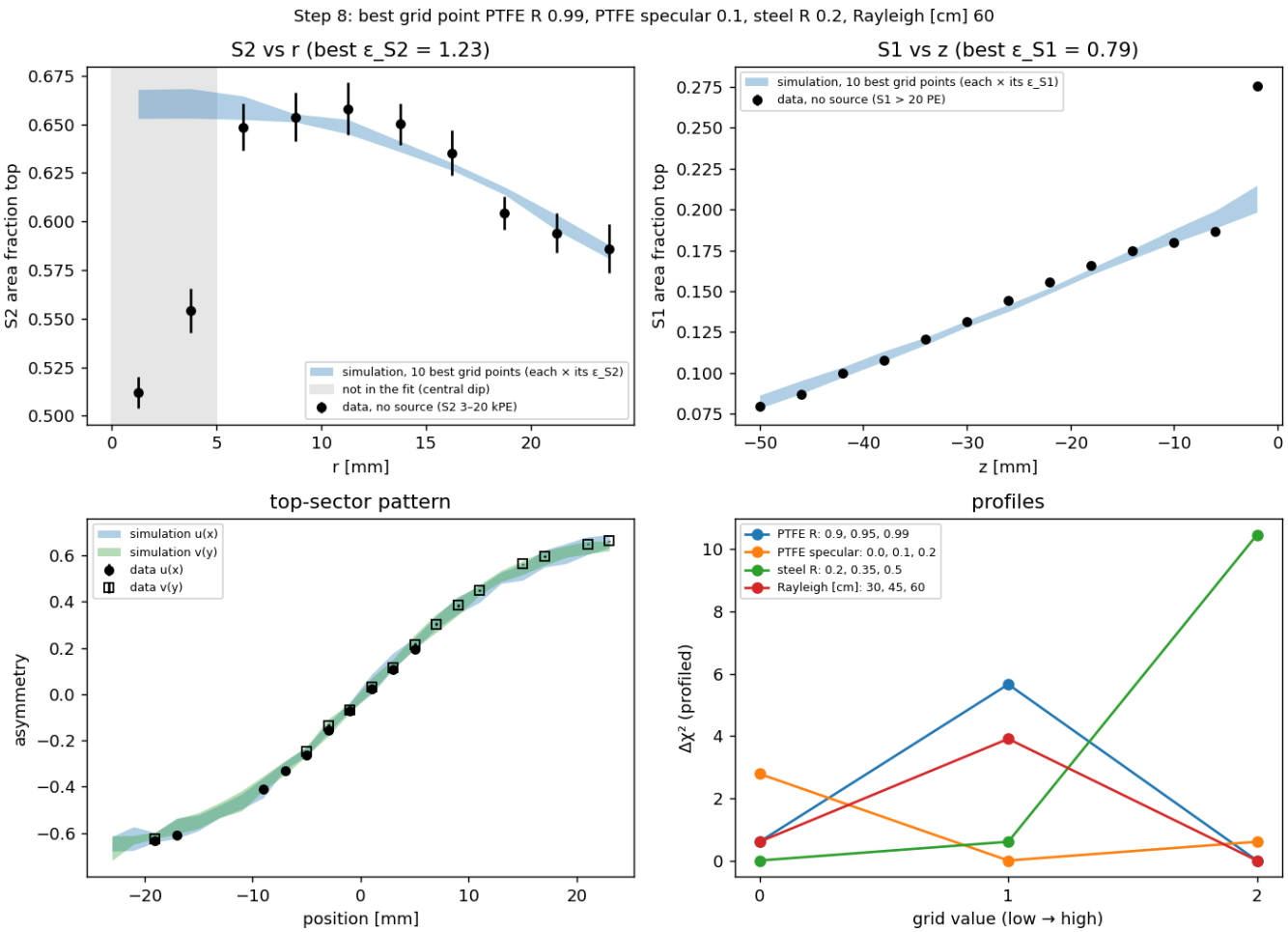


Top deficit or bottom excess? Cs-137 photopeak (fixed S2 per event, LED gains): top, bottom, total and area fraction top versus r, each ÷ its value at r = 7.5–12.5 mm; data points vs simulation band.

• **Sector pattern:** data inside the simulation band over ±24 mm, identical for all sources — geometry right.
• **S2 versus size and source:** no source 0.68 (3–5 kPE) → 0.53 (> 100 kPE); Na-22 0.50 → 0.38; Cs-137 0.37 → 0.28. Rate and size dependence are top-PMT effects (load, large pulses), not geometry. The simulation (0.55–0.60) agrees with the low-load, medium-size data.
• **S2 versus r:** outside the centre data/simulation top/bottom ratio 1.16–1.28 (absorbed by ε_{S2}); at r < 5 mm 0.65–0.77 (central dip in every source, also for single scatters).
• **S1 versus z:** same shape, ratio 0.70–0.77 at every depth (one efficiency factor for small signals on the top PMT).
• **Top or bottom?** Cs-137 photopeak (fixed S2 per event), single scatters: at the centre the area fraction top shows no dip (1.00 vs simulation 1.01–1.04) — the dip is a small-S2 effect. Outwards top and bottom both fall; the area fraction top at r = 24 mm is 0.90–0.92 (simulation 0.88–0.90), but the total S2 light drops to 0.44–0.51, far more than the optical 0.78: charge or S2 light is lost near the wall. (With all events the bottom seemed to rise outwards — that was a multiple-scatter effect.)
• Implied top-PMT efficiency relative to geometry (S2): ≈ 1 without source, ≈ 0.3 for Cs-137 large S2 — the v2 gain factor f_top (0.74–0.88, anchored on Cs runs) corrects only part of it.

5. Fit of the optical parameters

81-point grid (PTFE reflectivity 0.90/0.95/0.99 × PTFE specular fraction 0/0.1/0.2 × steel reflectivity 0.2/0.35/0.5 × Rayleigh length 30/45/60 cm) for S2 and S1, binned like the data. Per point two efficiency factors multiply the top/bottom ratio (ε_{S2}, ε_{S1}); the optical parameters are constrained by the shapes (S2 vs r for r ≥ 5 mm, S1 vs z, u/v), with a systematic floor on each observable.



Fit: simulation band = the 10 best grid points (each scaled with its own ε), data = points (no source); χ² profiles of the four optical parameters.

• Best point: PTFE R 0.99, specular 0.1, steel R 0.2, Rayleigh 60 cm; χ² 2.6 / 10.5 / 21.6 for 7 / 11 / 45 dof (S2 vs r / S1 vs z / u, v).
• ε_{S2} = 1.23 (1.14–1.29), ε_{S1} = 0.79 (0.71–0.80): not one QE ratio — with a QE ratio of ~1.2 the top PMT keeps only ~0.65 of its small S1 signals relative to the bottom, consistent with the known small-signal integration loss.
• Steel reflectivity low (0.2 preferred over 0.5, Δχ² 10.5); PTFE reflectivity, PTFE specular part and Rayleigh length are not constrained (Δχ² ≤ 6). (With all events the S2 shape fitted worse, χ² 11.2 vs 2.6.)
• Dead cross on the top PMT (1.5 / 3 / 6 mm): centre / ring at most 1.03 → 1.00, data 0.82 — not the cause of the central dip.
• Central dip versus S2 size (data, single scatters): centre / ring 0.69 (3–5 kPE) → 0.87 (20–50 kPE) → 1.02 (> 50 kPE), but mostly because the ring value falls with size (0.79 → 0.55) while the centre stays ≈ 0.48–0.54. A pure threshold loss from sharing the charge over four channels would make the centre rise with size; not seen. Likely a top-PMT signal-processing / per-channel effect rather than geometry (to check with fully integrated waveforms).

6. Design study: extra PTFE cylinders

PTFE cylinders with the radii of the drift-region PTFE, inside the ring bores, foil to foil:

extra PTFE	S2 light	S2 vs now	S1 light	S1 vs now	S2 top fraction
none (as built)	7.6 %	+0 %	7.6 %	+0 %	0.564
gate–anode	9.0 %	+19 %	8.2 %	+9 %	0.553
anode–top screen	8.7 %	+16 %	7.7 %	+2 %	0.596
cathode–bottom screen	7.9 %	+5 %	8.6 %	+13 %	0.540
gate–anode + anode–top screen	10.3 %	+37 %	8.5 %	+12 %	0.582
gate–anode + anode–top screen + cathode–bottom screen	10.8 %	+43 %	9.7 %	+28 %	0.559

Most rescued photons still end on meshes, PMT frames or in the photocathode (QE), so the gain is a factor ~1.4. Not studied: high voltage (PTFE across 5 kV gate–anode, charge-up in the extraction region), mechanics.

7. Anode mesh rotated by 120°

A square mesh repeats every 90°, so a 120° rotation of the anode relative to the gate is a 30° rotation. Simulated with the anode as now (aligned with the gate) and rotated ($4 \cdot 10^7$ photons each):

	anode as now	anode +120°
S2 top fraction	0.5631 ± 0.0003	0.5632 ± 0.0003
S2 light collection top / bottom	4.25 / 3.30 %	4.25 / 3.30 %
S2 light absorbed on the anode foil	12.3 %	12.3 %
S1 top fraction	0.1769	0.1767
fine structure of the top light vs S2 position (1 mm, beyond statistics)	3.3 %	2.8 %

Optically nothing changes (radial and depth profiles and the sector pattern coincide as well). Not covered: the electric field — a rotation changes where gate and anode bars cross, and with it the local extraction and electroluminescence field.

8. Limitations and next steps

• Everything outside |x|, |y| < 60 mm is black and the three gate–anode pillars are missing: ~22 % of the S2 light leaves the model sideways. The surrounding structure (holders, field-cage parts, cryostat wall) is next (TODO) — it affects light collection and the design study, little the top fraction.
• Central S2 dip: check the position reconstruction near u = v = 0 (AFT versus u, v directly), since the PMT cross is excluded.
• Optical parameters: a finer grid with more photons (u/v term) to constrain PTFE reflectivity and Rayleigh length; measure the top-PMT small-signal efficiency (amstrax integration) to separate it from the QE ratio.
• Oct 2024 (2 + 6 mm liquid/gas) vs now (4 + 4 mm): the simulation predicts only +0.01 in the S2 top fraction.